



INTRODUCTION TO PRODUCTION AND MANUFACTURING SYSTEMS

INDU 211

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Agenda

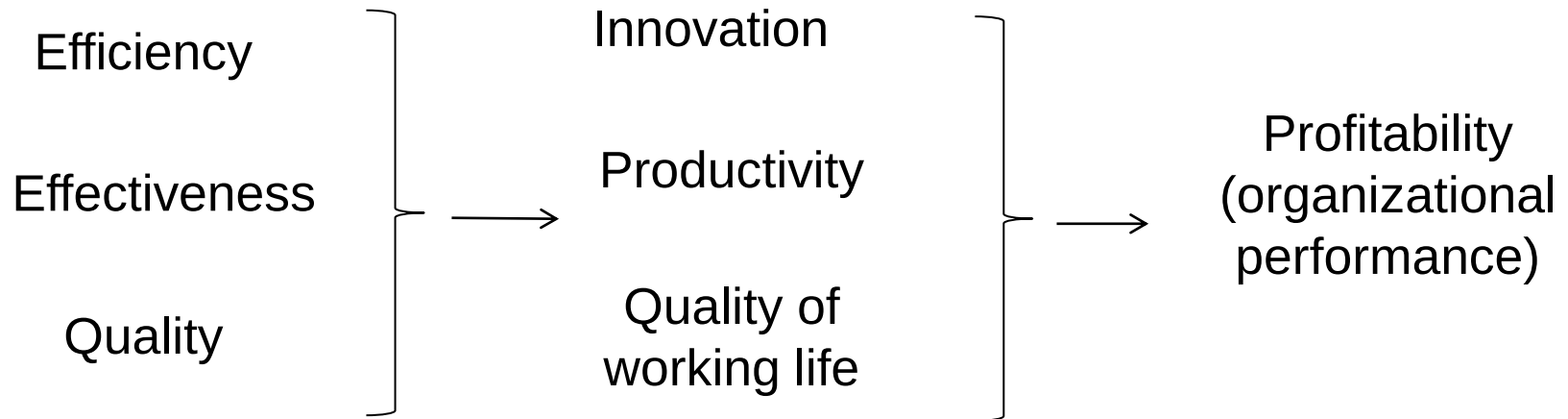
- Chapters 6 and 11 (Work Design and Human Factors)

Problematic

- To produce a given units of a product (10,000 units per year)
 - To manage labor and design jobs so people are effectively and efficiently utilized
 - Equipment requirement?
 - Labor requirement?
 - Design the most efficient work procedure
 - Training operators
 - Incentive system
 - Scheduling the orders received
 - How efficient the operations are performing?
 - How much to charge for the product?
 - Cost estimation
 - Measure productivity
- Time study

Organization performance

- The need for an efficient and effective control system



Productivity = output/input

- Work measurement: the standard by which we measure efficiency
- Work design: helps determine efficiency

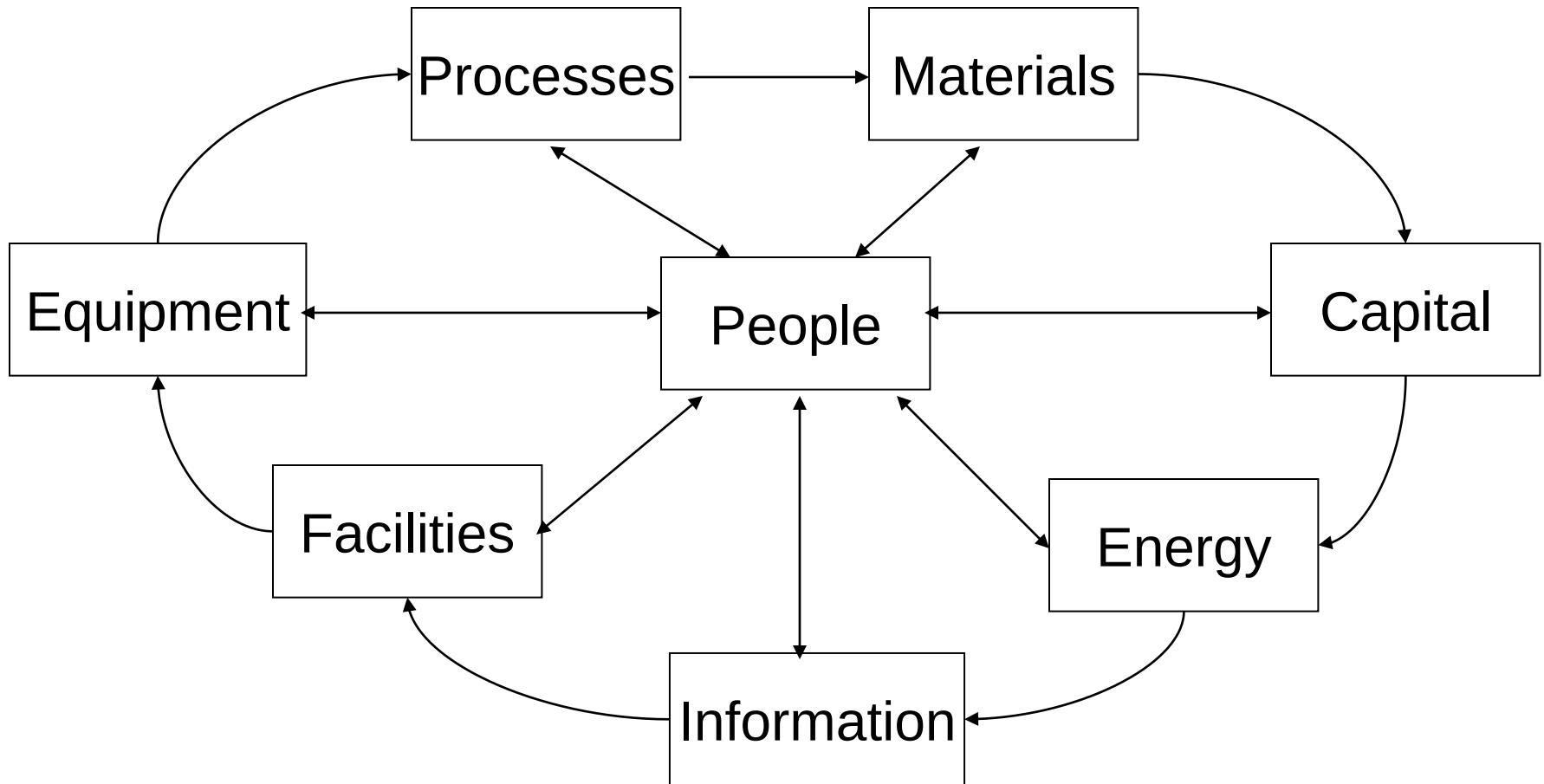
Measures of Productivity

Partial measures	$\frac{\text{Output}}{\text{Labor}}$	$\frac{\text{Output}}{\text{Machine}}$	$\frac{\text{Output}}{\text{Capital}}$	$\frac{\text{Output}}{\text{Energy}}$
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Multifactor measures	$\frac{\text{Output}}{\text{Labor} + \text{Machine}}$	$\frac{\text{Output}}{\text{Labor} + \text{Capital} + \text{Energy}}$
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Total measure	$\frac{\text{Goods or Services Produced}}{\text{All inputs used to produce them}}$
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Productive Work System



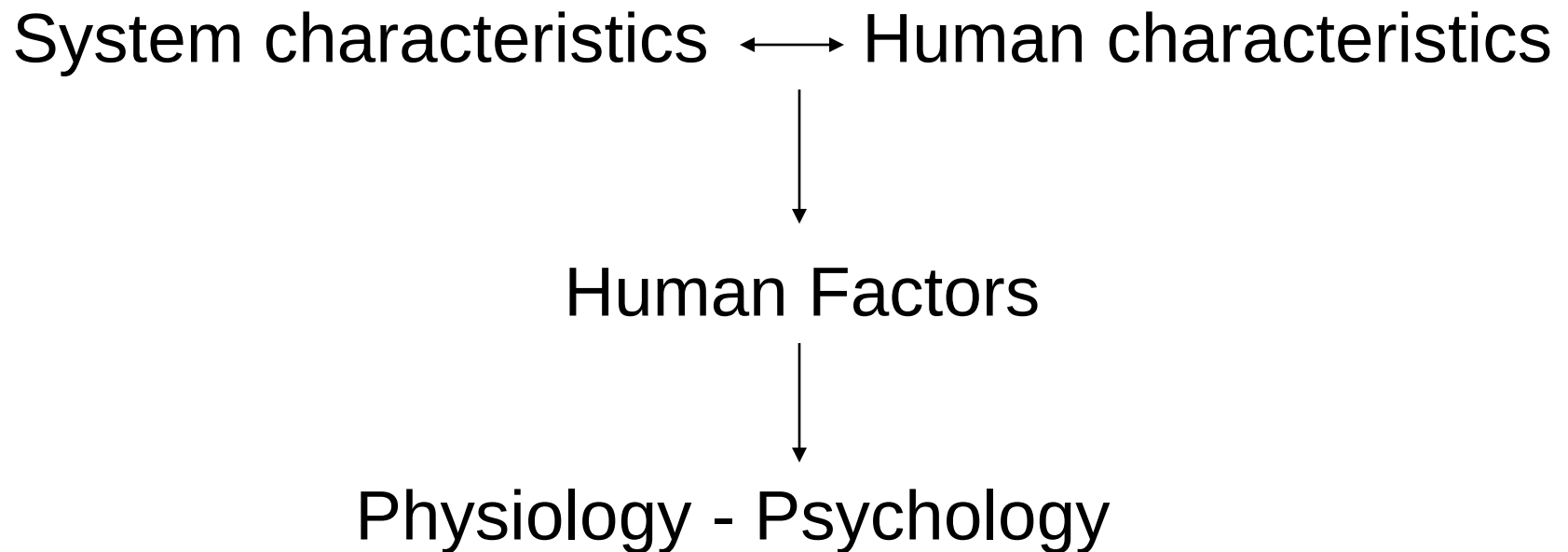
Strategy: Human Resources

- Culture of caring for people in the totality of their lives, not just at work
- To manage labor and design jobs so people are effectively and efficiently utilized

Human Factors Engineering

- International Ergonomics Association:
 - Ergonomics (or human factors) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system, and the profession that applies theory, principles, data, and other methods to design in order to optimize human well-being and overall system performance.

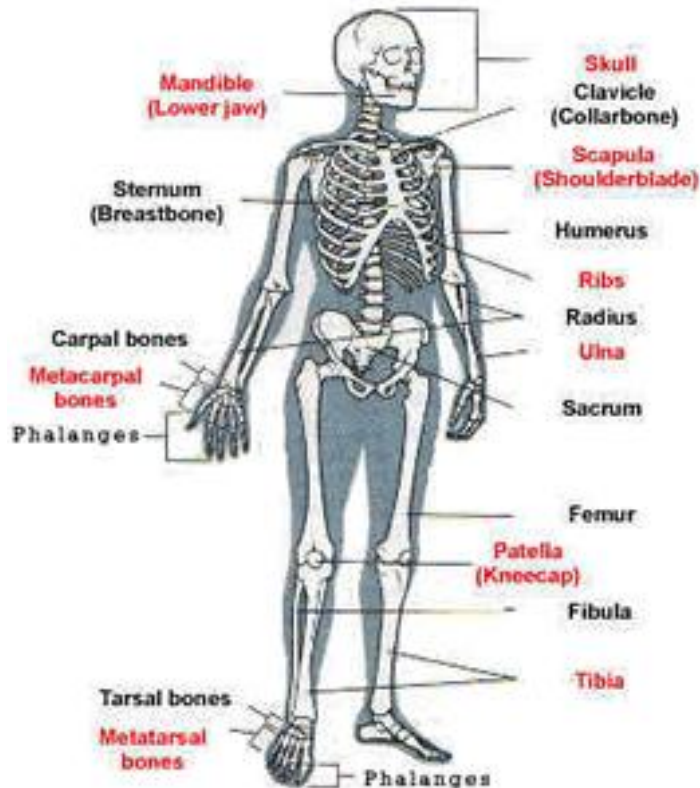
Human Factors



Psychology

- Humans...
 - Like money
 - Don't want to be bored
 - Don't always want to do a lot
 - Have brains
 - Deserve information, fair treatment, and most of all respect
 - and humans make mistakes

Physiology



- Biological characteristics of human beings
 - Structure of the skeleton
 - Skeletal muscles
 - Nervous system
 - Metabolic processes

Physiological vs psychological characteristics of human performance

- Physiological characteristics
 - Physical aspects of human activities
 - Lifting, reaching, carrying, hearing, seeing, speaking
- Psychological characteristics
 - Mental aspects of human activities
 - Stress, boredom, motivation

Machine vs. Human Characteristics

- Range of operations
- Size of workpiece
- Speed of operation
- Tolerance capabilities
- Energy consumption
- Maintenance requirements

Machine vs. Human Characteristics

<i>Performance Characteristic</i>	<i>"Typical" Machine Tool</i>	<i>"Typical" Human Being</i>
Range of Operations	Most M.T.'s perform one basic operation; some perform up to 10.	Extremely broad range.
Size of Workpiece	Some machines perform operations on microscopic-sized workpieces; others machine workpieces weighing several tons.	Very limited range. Cannot perform operations on extremely small items. Can perform repetitive operations on items weighing up to 50 pounds, where physical handling is required.
Speed of Operation	Can be extremely fast.	Usually very slow.
Tolerance Capabilities	Machines routinely hold tolerances to 0.001 inch.	Very poor at accuracy and repeatability.
Energy Consumption	Very efficient in converting energy into useful output.	Very low level of energy output; cannot sustain output on even level.
Maintenance Requirements	Regular maintenance required; complete overhaul needed periodically.	Needs minor "maintenance" frequently, such as water, food, oxygen, rest room breaks, bandaids, etc. Major overhauls are performed continuously, as the body mends itself.
Response to Unexpected Occurrences	Very limited capability.	Extremely resourceful and creative.

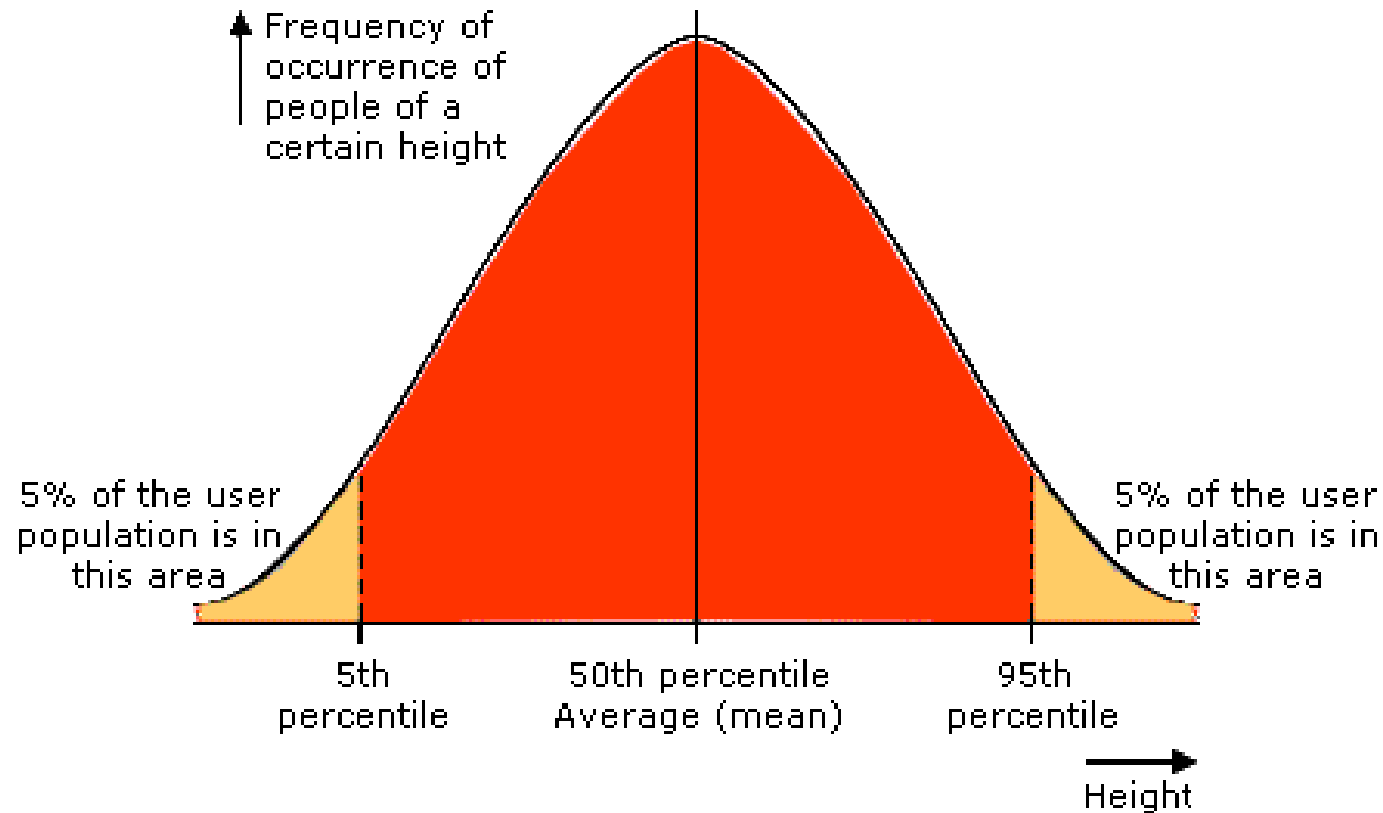
Anthropometry

- The scientific study of the measurements and proportions of the human body
- Deals with body shape and size:
 - People come in all shapes and sizes
 - Their physical characteristics must be taken into account in the design of something as simple as a pencil to something as complex as a car

Decide who you are designing for

- Need to know exactly *who* you are designing for
- Decide whether you are designing for the average or extremes
- Anthropometry tables give measurements of different body parts for men and women (according to nationalities, age groups, etc.)
- Decide which body measurements are relevant to your design

Decide who you are designing for



Decide who you are designing for

What is it that you are aiming for with your design?	Design examples:	Examples of measurements to consider:	Users that your design should accommodate:
Easy reach	Vehicle dashboards, Shelving	Arm length, Shoulder height	Smallest user: 5th percentile
Adequate clearance to avoid unwanted contact or trapping	Manholes, Cinema seats	Shoulder or hip width, Thigh length	Largest user: 95th percentile
A good match between the user and the product	Seats, Cycle helmets, Pushchairs	Knee-floor height, Head circumference, Weight	Maximum range: 5th to 95th percentile
A comfortable and safe posture	Lawnmowers, Monitor positions, Worksurface heights	Elbow height, Sitting eye height, Elbow height (sitting or standing?)	Maximum range: 5th to 95th percentile
Easy operation	Screw bottle tops, Door handles, Light switches	Grip strength, Hand width, Height	Smallest or weakest user: 5th percentile
To ensure that an item can't be reached or operated	Machine guarding mesh, Distance of railings from hazard	Finger width Arm length	Smallest user: 5th percentile Largest user: 95th percentile

Human Factors (ergonomics)

- The design of human-machine interface to improve efficiency and effectiveness
- Machine redesign to integrate human factors
- Design of workstations
- Design of tools
- Design of a cockpit for a modern jet



Human interface with the world of work

- Interfaces
 - Work environment
 - Physical environment
 - Social structures
 - Basic values, religion, laws, traditions, legal structure, educational system
 - Machines
 - Information/Communication systems
 - Organizational/supervisory structure
 - Robots

Interface with Work environment

- Industrial Engineers design workplaces
 - Combination of tables, desks, chairs, equipment, telephone, computer, and other devices
- Optimize work space for comfort, visibility, access, safety, enjoyment
 - Arrangements of workplace components (e.g., optimal location of dials, gauges, levers, bottoms in the workplace of a pilot)
 - Illumination
 - Atmospheric conditions (temperature, humidity, air clearance)

Interface with Machines

- Optimize for user comfort, visibility and access to controls
 - **Visibility** - Can people of different sizes see when they operate equipment or vehicle?
 - **Ingress and egress** - Can target population easily climb in and out of equipment or vehicles?
 - **Reaching and grasping** - Are controls placed so everyone can reach and operate them?
 - **Foot pedal operation** - Can people of various sizes comfortably operate foot pedals?
 - **User maintenance** - Can everyone perform maintenance tasks?
 - **Strength assessment** - Does operating the machine require inordinate strength or face the potential for injury?

Interface with Information/Communication Systems

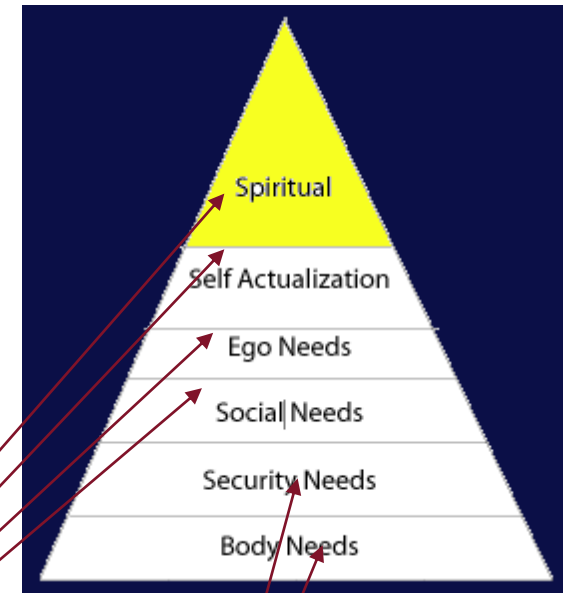
- Optimize for ease of access to, understanding of, communicating, presentation of information
- Written reports, information displays, phone calls, buzzers, lights, signals, etc.
- Auditory vs. visual presentation

Interface with Organizational/Supervisory Structure

- Optimize for job satisfaction and high morale and motivation
 - Job design
 - Reporting to boss
 - Being evaluated/rewarded
 - Training

Motivation and Money

- Taylor's scientific management (1911)
 - Workers are motivated mainly by money
 - Suggested piece-rate system (payment per unit)
- Maslow's theory (1943)
 - People are motivated by hierarchy of needs, which includes money
- Herzberg (1959)
 - Theory of motivation
 - Importance of job enrichment, job rotation, employee empowerment and recognition



Growth
needs

Basic
needs

Monetary Incentives

- Bonuses: Cash and stock options
- Profit sharing: Distribution of profits
- Incentive systems
 - Measured daywork: Pay based on standard time
 - Piece rate: Pay based on pieces done

Workforce planning

- To manage labor and design jobs so people are effectively and efficiently utilized
 - Equipment requirement
 - Labor requirement
 - Design the most efficient work procedure
 - Training operators
 - Incentive system
 - Scheduling the orders received
 - How efficient the operations are performing?
 - Cost estimation
 - Measure productivity
- Time study
- Learning curve



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Work Design

- Specifying the tasks that make up a job for an individual or group
- Involves determining
 - What is to be done (i.e., steps)
 - How it is to be done (i.e., tools etc.)
 - Why it is to be done (i.e., purpose)
 - Where is it to be done (i.e., location)
- Results in job description

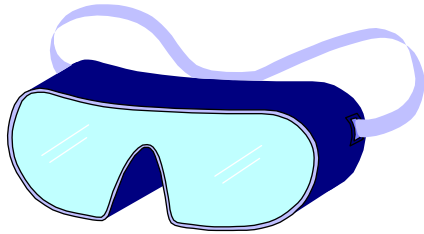
Work design components

- Working Conditions
- Job Specialization
- Behavioral Approaches to Job Design
- Methods Analysis and Improvement
- Work Measurement



Working Conditions

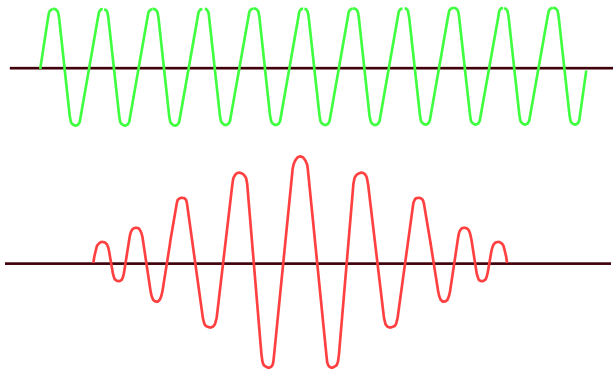
Safety and Security



Causes of Accidents



Noise and Vibration

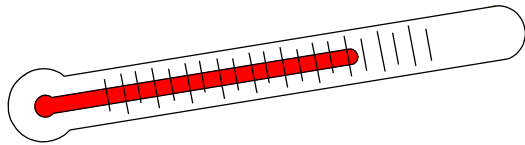


Work Breaks

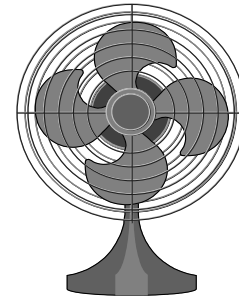


Working Conditions

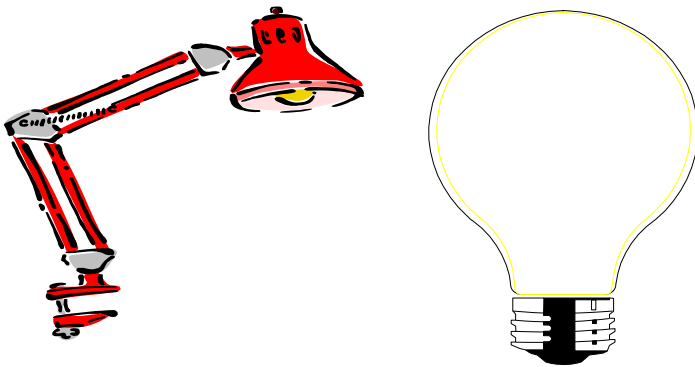
Temperature & Humidity



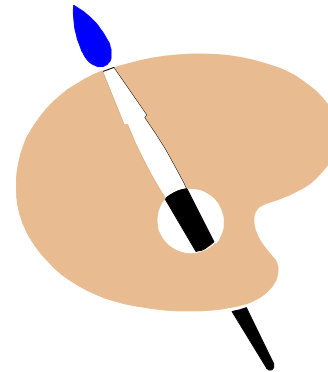
Ventilation



Illumination



Colour



Job Specialization

- Involves
 - Breaking jobs into small component parts
 - Assigning specialists to do each part
- First noted by Adam Smith (1776)
 - Observed how workers in pin factory divided tasks into smaller components
- Found in manufacturing & service industries

Specialization in Business: Advantages

For Management:

1. Simplifies training
2. High productivity
3. Low wage costs

For Labor:

1. Low education and skill requirements
2. Minimum responsibilities
3. Little mental effort needed

Disadvantages

For Management:

1. Difficult to motivate quality
2. Worker dissatisfaction, possibly resulting in absenteeism, high turnover, disruptive tactics, poor attention to quality

For Labor:

1. Monotonous work
2. Limited opportunities for advancement
3. Little control over work
4. Little opportunity for self-fulfillment

Behavioral Approaches to Job Design:

Job Expansion

- Process of adding more variety to jobs
- Intended to reduce boredom associated with labor specialization
- Methods
 - Job enlargement
 - Giving a worker a larger portion of the total task
 - Job enrichment
 - Increasing responsibility for planning and coordination tasks
 - Job rotation
 - Workers periodically exchange jobs
 - Employee empowerment

Methods Analysis and Improvement



- Focuses on how task is performed
- Used to analyze
 - Movement of body, people, or material
 - Activities of people and machines

Tools and Techniques

- Movement of individuals or materials (flow diagrams or process charts)
- Activity of human and machine (Gantt charts, activity charts, left-hand right-hand charts)
- Body movement (primarily arms and legs) (motion studies)
- Human Factors

Work method simplification (improvement)

- A systematic approach to finding easier and better ways to accomplish a task
- The basic objective: to avoid any kind of waste (“work smart not hard”)
- Essential step before time study
- Five-step procedure
 - Selecting a job
 - Getting and recording the facts
 - Questioning every detail
 - Developing and testing a better method
 - Installing and maintaining improvements
- Charts – steps 2 and 3

Flow process charts

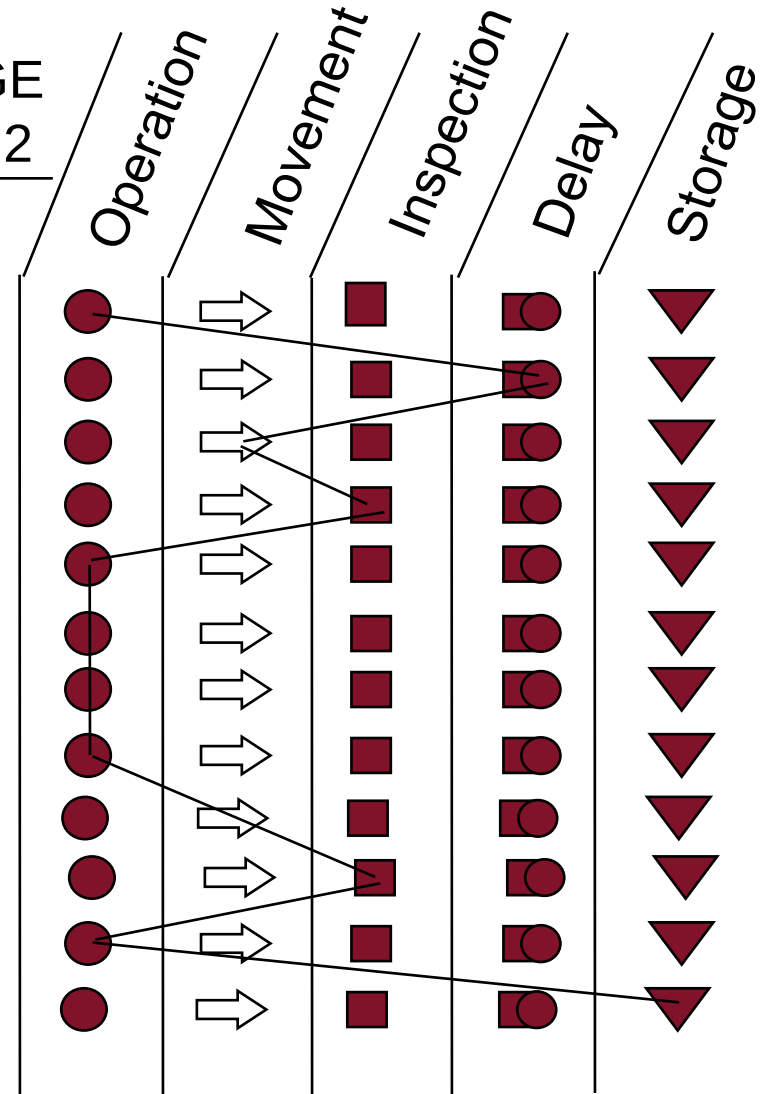
- A chart of all activities involved in a process
 - Operations, inspections, storage, transportation, and delays
- Questions to be asked:
 - Is this activity necessary or can be eliminated?
 - Can this activity be combined with others?
 - Is this the proper sequence of activities?
 - Can this activity be improved?
 - Is this the proper person to be doing this activity?

Process Flow Chart

Job: Requisition of petty cash	ANALYST D. Kolb	PAGE 1 of 2
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Details of Method

Requisition made by department head
 Put in “pick-up” basket
 To accounting department
 Account and signature verified
 Amount approved by treasurer
 Amount counted by cashier
 Amount recorded by bookkeeper
 Petty cash sealed in envelope
 Petty cash carried to department
 Petty cash checked against requisition
 Receipt signed
 Petty cash stored in safety box



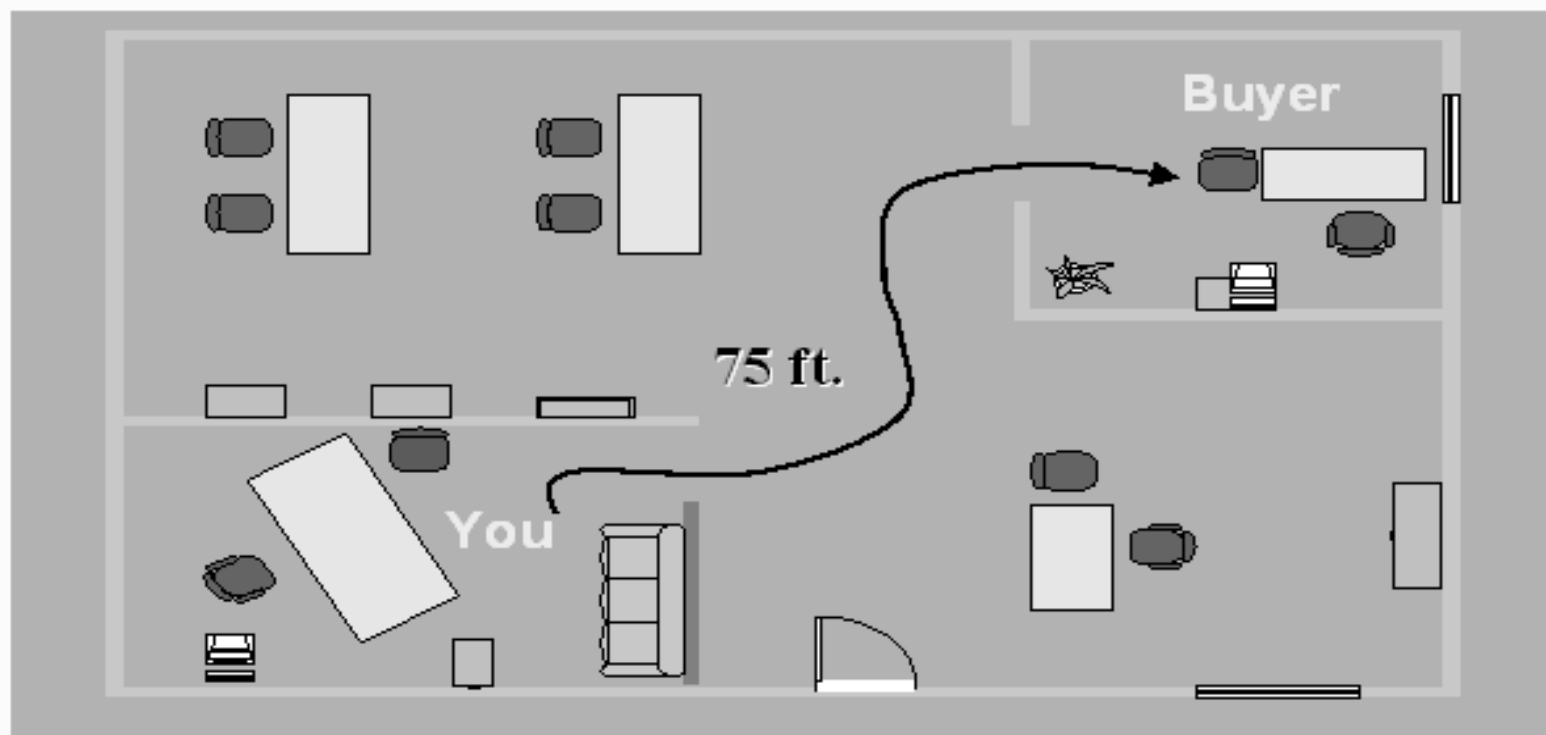
Process Chart

SUBJECT: Request tool purchase			
Dist (ft)	Time (min)	Symbol	Description
		○ ⇒ □ D ▽	Write order
		○ ⇒ □ D ▽	On desk
75		○ ⇒ □ D ▽	To buyer
		○ ⇒ □ D ▽	Examine

○ = operation; ⇒ = transport; □ = inspect; D = delay; ▽ = storage



Flow Diagram



Left-hand-Right-hand charts

- Useful in analyzing the work performed by one person at one specific work station
 - Follows the motion of the left and right hands of one operator
 - Useful for repetitive works

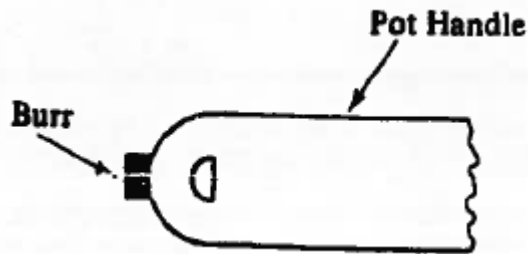
Operations Chart (Left Hand/Right Hand)

Left Hand		Right Hand	
Present Method	Symbol	Symbol	Present Method
1 Reach for bolt	○ ⇒ □ D ∇	○ ⇒ □ D ∇	Idle
2 Grasp bolt	○ ⇒ □ D ∇	○ ⇒ □ D ∇	Idle
3 Hold bolt	○ ⇒ □ D ∇	○ ⇒ □ D ∇	Reach for washer
4 Hold bolt	○ ⇒ □ D ∇	○ ⇒ □ D ∇	Assemble

○ = operation; ⇒ = transport; □ = inspect; D = delay; ∇ = storage



Example-current procedure



Plastic injection molding

Operation <u>Grinding Burrs</u>		Name <u>Bear Brian</u>		Date <u>7/30</u>	
Existing <u>X</u>		Proposed _____			

The diagram shows a top-down view of the grinding station. A central square represents the "Grinding Table". Inside it is a smaller square labeled "Grinding Wheel". To the right of the table is a vertical line labeled "Operator". To the left of the table is a rectangle labeled "Chute". To the right of the operator are three boxes: "Full Boxes" (top), "Completed Units" (middle), and "Empty Boxes" (bottom).

Left-hand Activity	Symbol	Symbol	Right-hand Activity
1. Reach to Chute	◇	□	1. Wait
2. Pick up 3 Parts	○	□	2. Wait
3. Move to Center	◇	□	3. Wait
4. Hold Parts	▽	○	4. Position Part and Grind
5. Hold Parts	▽	◇	5. Move to Right and Drop in Box
6. Hold Parts	▽	◇	6. Move Back to Center
7. Hold Parts	▽	○	7. Repeat 4
8. Hold Parts	▽	◇	8. Repeat 5
9. Hold Parts	▽	◇	9. Repeat 6
10. Hold Parts	▽	○	10. Repeat 4
11. Hold Parts	▽	◇	11. Repeat 5
Repeat Steps 1 through 8 56 times.			
9. Walk to Box	□	□	9. Same as left hand
10. Slide Full Box Out	◇	◇	10. Same as left hand
11. Move Empty Box In	◇	◇	11. Same as left hand
12. Walk Back	□	□	12. Same as left hand

Example- improved procedure

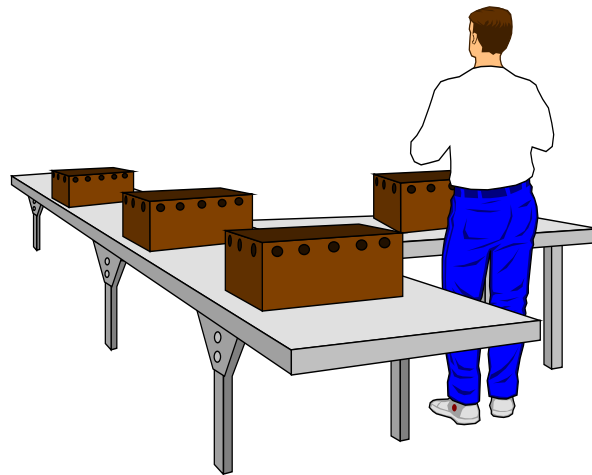
- 37% time reduction
- Significant material handling reduction

Operation <u>Grinding Burrs</u>		Name <u>Ara Smith</u>		Date <u>7/30</u>	
Existing <u> </u>		Proposed <u>X</u>			

Left-hand Activity				Right-hand Activity
1. Reach to and Preposition Handle	◇	◇	◇	1. Reach to and Preposition Handle
2. Slide to Grinder	◇	◇	◇	2. Slide to Grinder
3. Grind	○	○	○	3. Grind
4. Slide to Center and Drop	◇	◇	◇	4. Slide to Center and Drop

Motion study

- The systematic study of the human motions used to perform an operation
- Analysis of therbligs - basic elemental motions into which a job can be broken down
- Micro-motion study - use of motion pictures and slow motion to study motions that otherwise would be too rapid to analyze



Motion Study Techniques

- Guidelines for designing motion-efficient work procedures
- Eliminate
 - All possible jobs, steps, or motions
 - Use of hand as a holding device
 - Abnormal motions, use of muscles, danger
 - Idle time
- Combine
 - One continuous curved motion instead of short motions with sudden changes in direction
 - Tools, controls, motions
- Rearrange
 - Distribute the work evenly between the two hands
- Simplify
 - Reduce eye travel
 - Use momentum to build up energy
 - Shorten motions

Work Measurement

- Determines the amount of work time required to generate one unit of output
- Provides labor standards
 - Target amount of time required to perform a job under normal working conditions



Uses of Labor Standards

- Costing labor content of products
- Cost and time estimates for bids
- Planning staffing needs
- Planning production
- Wage-incentive plans
- Employee efficiency

Definitions

- Normal Time: The time required for an average trained operator to perform a task under usual working conditions and working at a normal pace.
- Normal Pace: The pace of an average, trained and conscientious operator working over an 8-hour day.

Definitions

- Actual Time: The observed time required for an operator to perform a task.
- Allowances: The amount of time added to the normal time to provide for personal needs, unavoidable delays, and fatigue.
- Performance Rating: The adjustment of actual (observed) time to obtain the normal time.
- Normal Time=Observed Time X Performance Rating

Developing a Time Standard through time study

- Observed (Actual) Time (OT) = $\Sigma x_i/n$
 - Σx_i = sum of recorded times
 - n = number of observations
- Normal Time (NT) = OT x PR
 - PR = performance rating
- Standard Time = NT x AF
 - AF: Allowance factor
 - Personal needs, fatigue, unavoidable delays (e.g., machine or tool break down)

Example 1- Direct time study

Observation i	Time x_i (minutes)
1	1.12
2	1.15
3	1.16
4	1.12
5	1.15
6	1.18
7	1.14
8	1.14
9	1.19
Total	10.35

- Performance rating is 1.13 (the analyst believes that the operator is performing 13% faster than would normally be expected)
- Allowance is 20% of job time. Find the standard time for this job.

Example 1- Direct time study

Actual time = $10.35/9 = 1.15$

Normal time:

$$1.15 * 1.13 = 1.3$$

Standard time:

$$1.3 * 1.2 = 1.56$$

Example 2

Element	Average time
1	15
2	7
3	5
4	8

Performance rating = 115% or 90%

Allowances = 12%

Total actual time = $15 + 7 + 5 + 8 = 35$

Normal time = $35 * 1.15 = 40.25$ or $35 * 0.9 = 28.5$

Standard time = $40.25 * 1.12 = 45.08$ or $28.5 * 1.12 = 31.92$